

Gundarapu Jaswanth Ganesh

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SUMMARY

Aspiring Software Developer with hands-on experience building Python-based backend systems, AI applications, and automation workflows. Strong foundation in Python, SQL, FastAPI, PostgreSQL, AI/LLM applications, and system integration

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Backend & Databases:** FastAPI, REST APIs, Pydantic, SQLAlchemy, PostgreSQL, Redis
- **AI & Tools:** RAG, LLMs, Agentic AI, LangChain, Git, GitHub, VS Code
- **Core:** OOP, DBMS, Computer Networks

PROJECTS

Synapse – AI Enterprise Knowledge Platform, Python | FastAPI | RAG | PostgreSQL

- Built an enterprise knowledge platform integrating multiple knowledge sources with secure authentication, RBAC, and a modular FastAPI architecture.
- Engineered hybrid retrieval using dense vector search and BM25 with metadata filtering and citation grounding, targeting improved retrieval precision from ~72% to ~90% on a 50+ query benchmark.
- Developed asynchronous document ingestion with metadata-driven indexing to support **10,000+ documents** and improve processing efficiency.

EventFlow – Scalable Event Management System, Python | FastAPI | PostgreSQL | Redis | JWT

- Built a production-oriented booking backend with JWT authentication, role-based access control, event lifecycle management, and transactional seat reservation.
- Implemented database-level concurrency controls and Redis caching, improving API throughput by ~40% and reducing response latency by ~30% under simulated concurrent workloads.
- Added automated registration, capacity validation, and reliable booking workflows to handle concurrent user requests.

Personal AI Assistant — Agentic AI & Task Automation, Python | LLMs | Agentic AI | APIs

- Built an AI-powered personal assistant that interprets user requests, selects appropriate tools, and executes multi-step tasks through integrated APIs and services.
- Implemented agent-based task execution and tool integration, achieving ~85% task completion across a 30-task evaluation set and reducing average execution time by ~25%.
- Integrated modular tools and external services to support extensible task automation and real-time information retrieval.

EDUCATION

Vishnu Institute of Technology,
B.Tech in Electronics & Communication Engineering.

2022 – 2026
Bhimavaram, India

CERTIFICATIONS

- Oracle Certified Foundations Associate - Agentic AI ☑
- HackerRank — Python ☑